ANSIBLE COMMANDS

- ·ansible-playbook --check <playbook.yml> : Runs a playbook in "check" mode, simulating the changes without actually applying them to the system.

  ansible-playbook --check site.yml

- ·ansible-playbook --diff <playbook.yml> : Runs a playbook and shows the differences between the previous and new configurations on the target systems.

  ansible-playbook --diff site.yml

- ·ansible-vault create <file> : Creates a new encrypted file using Ansible Vault.

  ansible-vault create secrets.yml

- ·ansible-vault decrypt <file> : Decrypts an encrypted file created with Ansible Vault.

  ansible-vault decrypt secrets.yml

- ·ansible-pull -U <repository_url> : Runs a playbook on a local system by pulling it from a Git repository.

  ansible-pull -U https://github.com/your-org/ansible-playbooks.git

DOCKER COMMANDS

- ·docker exec -it <container_id> <command> : Executes a command inside a running container interactively.

  docker exec -it mycontainer bash
- ·docker stop <container_id> : Stops a running container.
- ·docker start <container_id> : Starts a stopped container.
- ·docker restart <container_id> : Restarts a running or stopped container.
- ·docker rm <container_id> : Removes a stopped container from the system.
- ·docker rmi <image_id> : Removes a Docker image from the local system.
- ·docker logs <container_id> : Fetches the logs from a running or stopped container.
- ·docker network ls : Lists all Docker networks.
- ·docker volume ls : Lists all Docker volumes.
- ·docker info : Displays detailed system information about Docker.
- ·docker stats : Displays real-time statistics about running containers, including CPU, memory usage, and network activity.

GITHUB COMMANDS

- ·git config : Configures user information like name and email for commits.
- ·git init : Initializes a new Git repository in the current directory.
- ·git clone : Creates a local copy of a remote repository.
  git clone https://github.com/user/repository.git

- ·git branch : Lists, creates, or deletes branches in the repository.
  git branch # List branches
  git branch new-branch # Create a new branch
  git branch -d old-branch # Delete a branch
- ·git checkout : Switches to a different branch or commit.
  git checkout main
- ·git checkout -b : Creates a new branch and switches to it.
  git checkout -b feature-branch
- ·git merge : Combines changes from one branch into another.
  git merge feature-branch
- ·git status : Displays the state of the working directory and staging area.
- ·git add : Stages changes for commit.
  git add file.txt # Adds a specific file
  git add . # "." Adds all modified files

GITHUB COMMANDS

- ·git commit : Saves changes to the local repository with a message.
  git commit -m "Initial commit"
- ·git log : Shows the commit history.
- ·git diff : Displays differences between the working directory and the last commit.
- ·git push : Uploads local commits to a remote repository.
  git push origin main
- ·git pull : Fetches and integrates changes from a remote repository.
  git pull origin main
- ·git fetch : Retrieves updates from a remote repository without merging.
  git fetch origin
- ·git reset : Reverts changes to a previous commit.
  git reset --hard <commit_id>
- ·git revert : Creates a new commit that undoes a specific commit.
  git revert <commit_id>
- ·git stash : Temporarily saves uncommitted changes.
- ·git stash pop : Restores stashed changes.
- ·git remote : Manages remote repository connections.
  git remote -v # List remote repositories
  git remote add origin https://github.com/user/repo.git# Add a remote repository
- ·git tag : Marks specific commits with a tag.
  git tag v1.0
  git push origin v1.0

DOCKER COMMANDS

- ·docker --version : Displays the current version of Docker installed on your system.
- ·docker pull <image> : Downloads an image from Docker Hub (or another registry) to your local machine.

  docker pull ubuntu
- ·docker build -t <tag> <directory> : Builds a Docker image from a Dockerfile in the specified directory.

  docker build -t myapp:latest . (myapp is the name & latest is the tag)
- ·docker images : Lists all the Docker images available on your local machine.
- ·docker ps : Lists all the running Docker containers.
- ·docker ps -a : Lists all containers, including those that are stopped.
- ·docker run <image> : Runs a container from an image, with options like port mapping and volume mounting.

" docker run -d -p 8080:80 --name webserver nginx"

This runs an nginx container in detached mode (-d), mapping port 80 of the container to port 8080 on the host.

LINUX COMMANDS

- ·ls : Lists the files and directories in the current working directory.
- ·cd : Changes the current directory to the specified one.
- ·pwd : Prints the current working directory.
- ·mkdir : Creates a new directory.

  $ mkdir new_folder
- ·rmdir : Removes an empty directory.

  $ rmdir old_folder
- ·rm : Removes (deletes) files or directories.
- ·cp : Copies files or directories from one location to another.

  $ cp file.txt /home/user/Documents
- ·mv : Moves or renames files or directories.

  $ mv oldfile.txt newfile.txt
- ·cat : Displays the contents of a file.
- ·grep : Searches for a specified pattern within a file.

  $ grep "hello" file.txt
- ·echo : Prints text or variables to the terminal.

  $ echo "Hello, World!"

LINUX COMMANDS

- ·chmod : Changes the permissions of a file or directory.
  $ chmod 755 script.sh
- ·chown : Changes the owner and group of a file or directory.
  $ chown user1:usergroup file.txt
- ·ps : Displays a list of currently running processes.
- ·top : Displays real-time system processes and resource usage.
- ·df : Displays information about disk space usage on mounted filesystems.
- ·du : Displays the disk usage of files and directories.
- ·find : Searches for files or directories in a specified location.
  $ find /home/user -name "*.txt"
- ·man : Displays the manual page for a specified command.
- ·sudo : Executes a command with superuser (root) privileges.
- ·wget : Downloads files from the web.
  $ wget http://example.com/file.zip
- ·ssh : Connects to a remote machine via the SSH protocol.
  $ ssh user@192.168.1.1
- ·exit : Exits the current shell or terminal session.
- ·curl: Transfer data from or to a server.

KUBERNETES COMMANDS

- ·kubectl version : Displays the version of kubectl and the Kubernetes cluster.
- ·kubectl get pods : Lists all the pods running in the current namespace.
- ·kubectl get pods -w : To watch
- ·kubectl get services : Lists all the services in the current namespace.
- ·kubectl get nodes : Displays all nodes in the cluster.
- ·kubectl describe <resource> <name> : Displays detailed information about a resource (e.g., pod, service).

  kubectl describe pod mypod
- ·kubectl apply -f <file> : Applies a configuration file to create or update resources.

  kubectl apply -f deployment.yaml
- ·kubectl delete <resource> <name> : Deletes a resource by name.

  kubectl delete pod mypod
- ·kubectl logs <pod-name> : Retrieves the logs of a pod.

  kubectl logs mypod
- ·kubectl exec -it <pod-name> -- <command> : Executes a command inside a running pod.

  kubectl exec -it mypod – bash
- ·kubectl get deployments : Lists all the deployments in the current namespace.

TERRAFORM COMMANDS

1.terraform init

Purpose: Initializes your Terraform working directory. This command downloads the necessary provider plugins and sets up the backend (if configured).

2.terraform plan

Purpose: Generates and shows an execution plan. It previews the actions Terraform will take to change your infrastructure based on the configuration.

3.terraform apply

Purpose: Applies the changes required to reach the desired state of the configuration. It will prompt for confirmation before proceeding unless the -auto-approve flag is used.

4.terraform destroy

Purpose: Destroys the infrastructure managed by Terraform. It will prompt for confirmation before proceeding unless the -auto-approve flag is used.

5.terraform validate

Purpose: Validates the syntax and internal consistency of your Terraform configuration files.

6.terraform show

Purpose: Displays the current state or a saved state file.

7.terraform refresh

Purpose: Updates the state file with the most recent data about your infrastructure. This is useful if your infrastructure has been changed outside of Terraform.

8.terraform state

Purpose: Interacts with the Terraform state, allowing you to inspect, modify, or remove resources from the state file.

ANSIBLE COMMANDS

- ·ansible --version : Displays the version of Ansible installed on your system.
- ·ansible-playbook <playbook.yml> : Runs an Ansible playbook file (playbook.yml) to execute a series of tasks on remote systems.

  ansible-playbook site.yml
- ·ansible <host> -m ping :Pings a remote host to verify if Ansible can communicate with it.

  ansible server1 -m ping
- ·ansible <host> -a <command> : Executes a specific command on a remote host. For example, ansible <host> -a "uptime".

  ansible server1 -a "uptime"
- ·ansible-inventory --list : Displays the inventory in JSON format, showing all hosts and groups defined in your Ansible configuration.
- ·ansible-galaxy install <role> : Installs an Ansible role from Ansible Galaxy, which is a repository for sharing and downloading roles.

  ansible-galaxy install geerlingguy.apache

KUBERNETES COMMANDS

- ·kubectl create -f <file> : Creates a resource defined in the specified file.

  kubectl create -f service.yaml
- ·kubectl scale deployment <name> --replicas=<number> : Scales a deployment to the specified number of replicas.

  kubectl scale deployment myapp --replicas=3
- ·kubectl expose : Exposes a resource, such as a pod or deployment, as a service.

  kubectl expose deployment myapp --port=8080 --target-port=80
- ·kubectl get namespaces : Lists all the namespaces in the Kubernetes cluster.
- ·kubectl config get-contexts : Lists all contexts in the kubeconfig file.
- ·kubectl config use-context <context-name> : Switches to a specific context in the kubeconfig.
- ·kubectl port-forward <pod-name> <local-port>:<remote-port> : Forwards a local port to a port on a pod.

  kubectl port-forward mypod 8080:80
- ·kubectl top pods : Displays resource usage (CPU/Memory) of pods.
- ·kubectl apply -k <directory> : Applies a Kustomize configuration from the given directory.